Gradient Descent — Motivating the Algorithm

$\min: f(x)$, $f: \mathbb{R}^n \to \mathbb{R}$, convex.

Locally: Around the point x_t

$$f(x) \approx \boxed{f(x_t) + \nabla f(x_t)^T (x - x_t)}$$

how to choose.

$$x_t \longrightarrow x_{t+1} \quad : \quad x_{t+1} = x_t + \eta d$$

$$x = x_t + \eta d$$

$$f(x_t + \eta d) \approx f(x_t) + \nabla f(x_t)^T (\eta d)$$

choose d to make: $\nabla f(x_t)^T d$ as small as possible.

Want: $d = -\nabla f(x_t)$

Gradient Descent:

$$\boxed{x_{t+1} = x_t - \eta \nabla f(x_t)}$$

Another motivation:

$$f(x_{t+1}) \approx \boxed{f(x_t) + \nabla f(x_t)^T (x_{t+1} - x_t) + \frac{1}{2\eta} \|x_{t+1} - x_t\|_2^2}$$

Minimize RHS by taking deriv (grad) & set $= 0$:

$$0 + \nabla f(x_t) + \frac{1}{\eta}(x_{t+1} - x_t) = 0$$

$$x_{t+1} = x_t - \eta \nabla f(x_t). \quad // \text{ GD}.$$

Important Questions

1. Is GD guaranteed to converge?

   Diverge?

2. How to choose step size ? ?

3. How fast does GD converge?

Choosing the Step Size

GD: $x_{t+1} = x_t - \eta_t \nabla f(x_t)$

- Fixed:

  $\eta_t = \eta \quad \forall t$

- Decreasing step size:

  $\eta_t \to 0$

  $\sum \eta_t = +\infty$

  Ex: $\eta_t = 1/t$ or $1/\sqrt{t}$

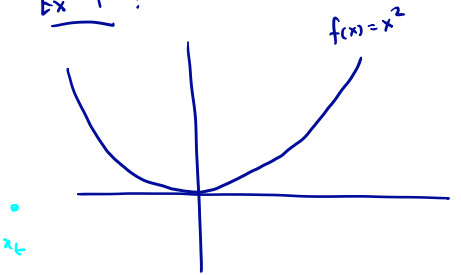
- Line search

Converge/Diverge? How Fast?

For now: fix $\eta_t = \eta$.

GD: $x_{t+1} = x_t - \eta \nabla f(x_t)$

Ex 1:

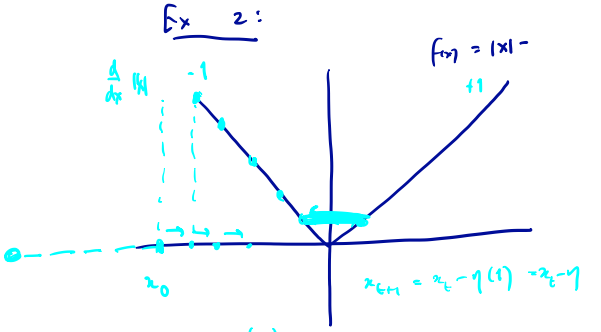


$x_{t+1} = x_t - 2\eta x_t$

very big if we are far away.
goes to zero as $x_t \to 0$

self-tuning

Ex 2:



$x_1 = x_0 - \eta(-1)$
$= x_0 + \eta.$